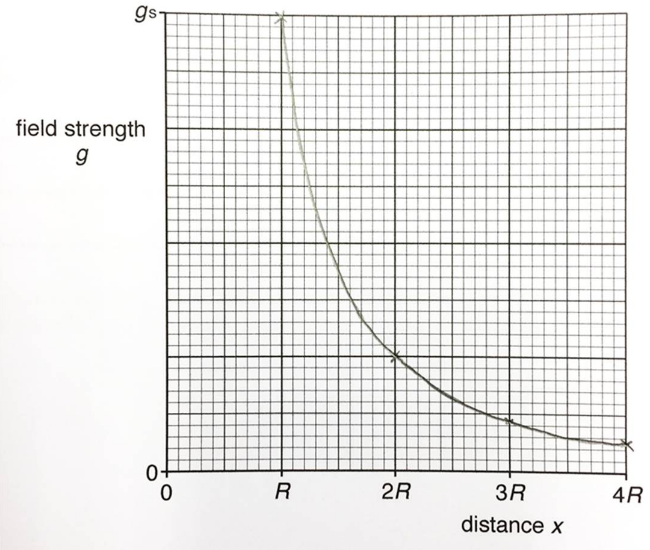
	be equal to that of a photon in free space.
9(a)(i)	$g_P = \dfrac{GM}{x^2}$ (magnitude)

9(a)(ii)	$E_P = -\dfrac{GMm}{x}$
9(b)(i)	
9(b)(ii)	Gravitational field strength is directly proportional to the mass of the sphere. A lower mass will result in all the values of field strength being lowered for the same value of x.
9(c)(i)	The centripetal force experienced by each star is the resultant force acting on each star. This resultant force is the gravitational force of attraction between the two stars, which is the same in magnitude for each of them, according to Newton's law of gravitation.
9(c)(ii)	$\omega = \dfrac{2\pi}{T}$ $\omega = \dfrac{2\pi}{4.0 \times 365 \times 24 \times 3600}$ $\omega \approx 4.98 \times 10^{-8}$ rad s^{-1}
9(c)(iii)	The gravitational force acting on star A provides the centripetal force necessary for start A to move in a circular motion. $F_g = M_A r \omega^2$ $\dfrac{GM_A M_B}{d^2} = M_A r \omega^2$ ----- (1) For star B, $\dfrac{GM_A M_B}{d^2} = M_B \left(d - r\right)\omega^2$ ----- (2) Solving (1) and (2), we have $1 = \dfrac{M_A r}{M_B \left(d - r\right)}$

	$1 = 3.0\dfrac{r}{\left(3.0\times10^{11}-r\right)}$ $r = 7.5\times10^{10}$ m
9(d)	From (1), we have $\dfrac{GM_AM_B}{d^2} = M_Ar\omega^2$. $\dfrac{6.67\times10^{-11}\times M_B}{\left(3.0\times10^{11}\right)^2} = 7.5\times10^{10}\times\left(4.98\times10^{-8}\right)^2$ $M_B \approx 2.51\times10^{29}$ kg Since $\dfrac{M_A}{M_B} = 3.0$, $\dfrac{M_A}{2.51\times10^{29}} = 3.0$ $M_A \approx 7.53\times10^{29}$ kg
9(e)	Half a period can be calculated by observing the stars when they overlap, and the next instant when they overlap.